

NoiPA / *Data Quality Atelier*

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I The Dataset

SELECT OR UPLOAD A SOURCE CSV

This atelier transforms raw CSV files from the Public Administration into refined quality reports. Select an existing dataset or upload a new one.

UPLOAD

OR SELECT AN EXISTING DATASET

⬆️ UPLOAD

200MB per file • CSV

attivazioniCessazioni.csv • 3234 KB

SELECTED
attivazioniCessazioni.csv

ROWS • COLUMNS
20,102 × 19

ROWS	COLUMNS	FILE SIZE	PREVIEW
20,102	19	3234.9 KB	200 ROWS

PREVIEW (FIRST 200 ROWS)

	_id	mese	anno	codice_ente	descrizione_ente	provincia_sede	regione_sede	attivazioni
0	6852caf7205349164bebcd69	11	2023	19	MINISTERO UNIVERSITA' E RICERCA	TO	1	250
1	6852caff6555da0907a40857	7	2023	2020	MINISTERO DELL'INTERNO - DIPARTIMENTO DELLA P.S.	BT	15	0
2	6852caef205349164bea2a6e	8	2021	2020	MINISTERO DELL'INTERNO - DIPARTIMENTO DELLA P.S.	Aq	12	0
3	6852cb029952cb647ef389e4	4	2023	2020	MINISTERO DELL'INTERNO - DIPARTIMENTO DELLA P.S.	AQ	12	0
4	6852cae9205349164be90859	6	2023	22	AGENZIA DELLE ENTRATE	VR	4	0
5	6852caed205349164be9b770	11	2023	2020	MINISTERO DELL'INTERNO - DIPARTIMENTO DELLA P.S.	SA	14	0
6	6852cae8205349164be8daf2	2	2024	12	MINISTERO DELL'INTERNO	TR	9	1
7	6852caef205349164bea272b	6	2023	22	AGENZIA DELLE ENTRATE	RG	18	6
8	6852caea205349164be90e4a	8	2023	954	ISPETTORATO NAZIONALE DEL LAVORO	BG	None	0

II The Pipeline

ELEVEN STAGES WORKING IN CONCERT

01 • SCHEMA	02 • COMPLETENESS	03 • CONSISTENCY	04 • ANOMALY	05 • CROSS-COLUMN	06 • DUPLICATES
07 • REMEDIATE	08 • GENERATE	09 • APPLY	10 • VERIFY	11 • NARRATIVE	

STAGE 11 OF 11 • COMPLETE 100%

PIPELINE COMPLETE ✓

RESET

SCHEMA ISSUES 13	MISSING COLS 11	FORMAT ISSUES 6	ANOMALIES 0	CROSS-COLUMN 8	DUP GROUPS 48
APPLIED 34	DEFERRED 56	FAILED 0	NOT NEEDED 9		

NARRATIVE REPORT

ACTION LEDGER

CLEANED DATASET

FINAL REPORT

Quality Report for attivazioniCessazioni

This quality report covers the attivazioniCessazioni dataset, with 20,102 cleaned rows processed in the remediation workflow. Overall quality posture is moderate: the validation run identified 86 section-level findings and signals, but no anomalies were detected. The main issue categories were schema issues, completeness gaps, consistency findings, cross-column findings, and duplicate groups. Specifically, the validation summary recorded 13 schema issues, 11 columns with missing values, 6 consistency findings, 8 cross-column findings, and 48 duplicate groups. No anomaly findings were reported, which reduces concern about extreme outliers or unusual records. The remediation process applied 34 actions, deferred 56 proposed actions, recorded 9 actions as not needed, and had 0 failed actions. Six generated cleaners were created to support the cleanup process. Verification confirmed 6 resolved items, including mese, anno, attivazioni, cessazioni, RATA→rata, and aggregation-time→aggregation_time. Residual risk is low based on the available briefing, with no unresolved risks reported, but deferred actions and remaining data quality findings indicate that some issues still require steward review.

Dataset Overview

The **attivazioniCessazioni** dataset contains **20,102 cleaned rows** and presents an overall quality posture that is **substantially improved but still not fully normalized**. The validation process identified **86 section-level findings/signals**, indicating that the dataset required broad review across schema, completeness, consistency, and cross-column behavior. In response, **34 remediation actions were applied**, **56 proposed actions were deferred without auto-apply**, and **0 actions failed**. In addition, **9 items were marked not needed**, which suggests that some potential interventions were assessed but ultimately not required.

The validation summary shows a mixed profile of issues rather than a single dominant defect type. There were **13 schema issues**, **11 columns with missingness**, **6 consistency findings**, **8 cross-column findings**, and **48 duplicate groups**, while **0 anomaly findings** were recorded. This pattern indicates that the dataset is structurally usable, but attention is still warranted for naming, completeness, and relational coherence across fields. The verification summary confirms that **6 issues were resolved**, specifically involving **mese**, **anno**, **attivazioni**, **cessazioni**, **RATA→rata**, and **aggregation-time→aggregation_time**. These resolved items demonstrate that the applied remediation improved both standardization and interpretability.

Overall, the dataset should be considered **operationally usable with residual quality debt**. The combination of successful fixes, zero failed actions, and a moderate number of deferred recommendations suggests a controlled cleanup process with clear remaining opportunities for further refinement.

Schema Validation

The schema validation step for **attivazioniCessazioni** focused on bringing column names into lowercase snake_case and aligning data types with the inferred schema. Several columns required renaming because their original labels contained characters or formatting that violate the naming rule.

COLUMN RENAMES

ORIGINAL NAME	NEW NAME	REASON
3descrizione	descrizione_3	Column name contains a leading digit, which violates the lowercase snake_case naming rule.
aggregation-time	aggregation_time	Column name contains a hyphen, which violates the lowercase snake_case naming rule.
att ivazioni	att_ivazioni	Column name contains whitespace, which violates the lowercase snake_case naming rule.
CODICE ENTE	codice_ente_2	Column name contains uppercase letters, and whitespace, which violates the lowercase snake_case naming rule.
Provincia Sede	provincia_sede_2	Column name contains uppercase letters, and whitespace, which violates the lowercase snake_case naming rule.
RATA	rata	Column name contains uppercase letters, which violates the lowercase snake_case naming rule.
regione%sede	regione_sede_2	Column name contains a percent sign, which violates the lowercase snake_case naming rule.

TYPE CASTS

COLUMN	ASSIGNED TYPE	NON-NULL COUNT
aggregation_time	datetime64[ns]	20102
anno	Int64	20102
att_ivazioni	Int64	19499
attivazioni	Int64	19900
cessazioni	Int64	19900
codice_ente	Int64	19499
descrizione_3	string	20102
descrizione_ente	string	19100
fonte_dato	string	160
_id	string	20102
mese	Int64	20094
note	string	300
provincia_sede	string	18936
qualifica	string	15016
rata	Int64	20102
regione_sede	Int64	19294

The casts show a consistent separation between temporal, numeric, and text fields. In particular, `aggregation_time` was standardized to `datetime64[ns]`, while core measure and identifier-like fields such as `anno`, `mese`, `attivazioni`, `cessazioni`, `codice_ente`, and `rata` were assigned `Int64`. Textual descriptors and metadata fields, including `_id`, `descrizione_3`, `descrizione_ente`, `fonte_dato`, `note`, `provincia_sede`, and `qualifica`, were cast to `string`. The non-null counts indicate that some descriptive columns are substantially sparser than the main numeric fields, but the schema now reflects a cleaner and more consistent structure for downstream processing.

Completeness Analysis

The completeness profile for **attivazioniCessazioni** shows a clear split between operational measures that are almost fully populated and descriptive fields that contain substantial placeholder or blank content. The table below summarizes the missing-like findings.

COLUMN	MISSING-LIKE COUNT	COMPLETENESS %	EXAMPLE TOKENS
codice_ente	546	97.2838523529997%	//, -, ?, ``, n.d.
descrizione_ente	1094	94.55775544721918%	`, ?, n.d., //, -
provincia_sede	1258	93.74191622724108%	?, ``, //, -, unknown
regione_sede	725	96.39339369216994%	`, ?, n.d., -, //
attivazioni	195	99.02994726892847%	N.D.
cessazioni	195	99.02994726892847%	N.D.
qualifica	5086	74.69903492189832%	`
note	19802	1.492388817033131%	`
fonte_dato	19942	0.7959407024176699%	`
Provincia Sede	367	98.17431101382948%	`
att ivazioni	195	99.02994726892847%	N.D.

A few patterns stand out. First, the numeric activity fields **attivazioni** and **cessazioni** are highly complete, each at **99.02994726892847%**, with **N.D.** appearing as the main missing-like token. Second, the location and entity descriptors—**codice_ente**, **descrizione_ente**, **provincia_sede**, and **regione_sede**—remain mostly complete but still show recurring placeholders such as empty strings, **?**, **n.d.**, **//**, and **-**. Third, the weakest fields are **note** and **fonte_dato**, which are almost entirely missing-like, indicating that these columns contribute very little usable content in the current extract. The duplicated or variant field names, such as **Provincia Sede** and **att ivazioni**, also suggest inconsistent naming or formatting in the source, even though their completeness is relatively high.

Format Consistency

The format review for **attivazioniCessazioni** shows that all six cleaned columns were successfully normalized, with no unresolved format issues remaining after verification.

MESE

- Expected Pattern:** Numeric month values in a consistent month format.
- Inconsistent Rows:** 744 rows were changed during cleaning.
- Examples of bad values:** '7', '8', '4', '6'.
- Transformation applied:** Applied successfully to standardize the month representation without altering the underlying values.
- Clean example when available:** '7'->'7', '8'->'8', '4'->'4', '6'->'6'.
- Outcome:** The column is now format-consistent and verified as resolved.

ANNO

- Expected Pattern:** Four-digit year values.
- Inconsistent Rows:** 585 rows were changed during cleaning.
- Examples of bad values:** '2023', '2021', '2024', '2022'.
- Transformation applied:** Applied successfully to preserve year values in a consistent year format.
- Clean example when available:** '2023'->'2023', '2021'->'2021', '2024'->'2024', '2022'->'2022'.
- Outcome:** The column is now format-consistent and verified as resolved.

ATTIVAZIONI

- Expected Pattern:** Numeric count values.

- **Inconsistent Rows:** 609 rows were changed during cleaning.
- **Examples of bad values:** '250', '0', '1', '6'.
- **Transformation applied:** Applied successfully to normalize the count field while keeping the original numeric content intact.
- **Clean example when available:** '250'->'250', '0'->'0', '1'->'1', '6'->'6'.
- **Outcome:** The column is now format-consistent and verified as resolved.

CESSAZIONI

- **Expected Pattern:** Numeric count values.
- **Inconsistent Rows:** 608 rows were changed during cleaning.
- **Examples of bad values:** '40', '2', '4', '3'.
- **Transformation applied:** Applied successfully to normalize the count field while preserving the original values.
- **Clean example when available:** '40'->'40', '2'->'2', '4'->'4', '3'->'3'.
- **Outcome:** The column is now format-consistent and verified as resolved.

RATA

- **Expected Pattern:** A compact period code in YYYYMM format.
- **Inconsistent Rows:** 802 rows were changed during cleaning.
- **Examples of bad values:** '202311', '202307', '202308', '202304'.
- **Transformation applied:** Applied successfully and the column was renamed to rata as part of the cleanup.
- **Clean example when available:** '202311'->'202311', '202307'->'202307', '202308'->'202308', '202304'->'202304'.
- **Outcome:** The column is now format-consistent under the cleaned name rata and verified as resolved.

AGGREGATION_TIME

- **Expected Pattern:** A timestamp/date-time value in a consistent standardized format.
- **Inconsistent Rows:** 1,607 rows were changed during cleaning.
- **Examples of bad values:** '2025-06-18T16:15:20.148346', '18/06/2025', '18.06.2025', 'GIU 18 2025'.
- **Transformation applied:** Applied successfully and the column was renamed to aggregation_time as part of the cleanup.
- **Clean example when available:** '2025-06-18T16:15:20.148346'->'2025-06-18T16:15:20.148346', '18/06/2025'->'2025-06-18T00:00:00.000000', '18.06.2025'->'2025-06-18T00:00:00.000000', 'GIU 18 2025'->'2025-06-18T00:00:00.000000'.
- **Outcome:** The column is now format-consistent and verified as resolved, with heterogeneous date representations consolidated into a single standard.

Anomaly Detection

Overview

No anomaly findings were recorded for the **attivazioniCessazioni** dataset. The available briefing does not report any numeric outliers, rare categories, or other irregular patterns requiring escalation. As a result, this section documents the absence of detected anomalies rather than describing specific exceptions.

Findings

- **Numeric outliers:** None were recorded.
- **Rare categories:** None were recorded.
- **Other anomaly types:** No additional anomaly findings were reported.

Interpretation

The lack of recorded anomalies suggests that the dataset did not present any values or categories that stood out during the anomaly detection review. In practical terms, this means there is no evidence in the provided material of extreme numeric values, unusual category frequencies, or isolated records that would warrant immediate correction or manual investigation.

Conclusion

Based on the supplied briefing, the anomaly detection outcome for **attivazioniCessazioni** is clean: no anomalies were identified and no remediation actions are indicated from this review alone.

Cross-Column Checks

NEAR-DUPLICATE COLUMN PAIRS

The dataset **attivazioniCessazioni** contains several column pairs that are almost identical in content, suggesting duplicated or inconsistently named fields that should be reconciled.

- `attivazioni` vs `att ivazioni`: 19,907 comparable non-missing rows, with 19,869 matches (**99.81% similarity**) and 38 mismatches.
- `codice_ente` vs `CODICE ENTE`: 19,556 comparable non-missing rows, with 19,499 matches (**99.71% similarity**) and 57 mismatches.
- `descrizione_ente` vs `3descrizione`: 19,008 comparable non-missing rows, with 18,899 matches (**99.43% similarity**) and 109 mismatches.
- `provincia_sede` vs `Provincia Sede`: 18,842 comparable non-missing rows, with 18,737 matches (**99.44% similarity**) and 105 mismatches.

These pairs are highly aligned overall, but the remaining mismatches indicate that the duplicated fields are not perfectly interchangeable.

SEMANTIC CONFLICTS ON NORMALIZED DUPLICATES

Some columns normalize to the same schema name but still disagree on a non-trivial number of rows, which is a stronger sign of cross-column inconsistency.

- `regione_sede` vs `regione%sede`: 9,708 rows where both values are present disagree after normalization, indicating a high-severity semantic conflict.
- `provincia_sede` vs `Provincia Sede`: 105 conflicting rows where both values are present.
- `codice_ente` vs `CODICE ENTE`: 57 conflicting rows where both values are present.

PERIOD CONSISTENCY CHECK

A cross-field validation also found a mismatch between the period indicator and the calendar fields:

- `anno`, `mese`, and `RATA`: 1,669 comparable rows show disagreement between the period column `RATA` and the year/month fields `anno` and `mese`.

OVERALL INTERPRETATION

The checks show a pattern of near-duplicate columns with very high agreement, alongside a smaller but important set of direct conflicts. The most critical issue is the disagreement between normalized duplicate fields and the period mismatch involving `RATA`, since these affect internal consistency across multiple columns rather than isolated value differences.

Row Duplicate Analysis

The duplicate review for **attivazioniCessazioni** identified both exact row duplicates and a smaller set of potential near-duplicates. The exact duplicates are defined by full-row matches after whitespace/case normalization, using the same broad set of columns including `_id`, `mese`, `anno`, `codice_ente`, `descrizione_ente`, `provincia_sede`, `regione_sede`, `attivazioni`, `cessazioni`, `RATA`, `aggregation-time`, `qualifica`, `note`, `fonte_dato`, `Provincia Sede`, `CODICE ENTE`, `3descrizione`, `regione%sede`, and `att ivazioni`. This indicates that the repeated records are not limited to a single business field, but span the complete observed row content.

Representative duplicate groups include:

- `[255, 1709]` — exact duplicate after normalization
- `[290, 19747]` — exact duplicate after normalization
- `[307, 11722]` — exact duplicate after normalization
- `[714, 12396]` — exact duplicate after normalization

- [925, 17914] — exact duplicate after normalization
- [1105, 16882] — exact duplicate after normalization

A second pattern appears in the near-duplicate findings, where rows share the inferred key columns ['_id', 'mese', 'anno', 'codice_ente'] but differ elsewhere. These cases suggest repeated entities or time-period records that are not fully identical and therefore require closer business-rule review rather than automatic deduplication.

Representative near-duplicate groups include:

- [160, 7699] — shared inferred key columns, differing in other fields
- [841, 15282] — shared inferred key columns, differing in other fields
- [1116, 9181] — shared inferred key columns, differing in other fields
- [2011, 7734] — shared inferred key columns, differing in other fields
- [2343, 13227] — shared inferred key columns, differing in other fields

Overall, the duplicate profile shows clear evidence of repeated rows at the full-record level, alongside a separate set of records that may reflect legitimate variation under the same inferred key.

Remediation Action Summary

Applied 34 remediation actions, with 56 deferred, 0 failed, and 9 not needed. The dataset **attivazioniCessazioni** was primarily improved through schema normalization, column renaming, type casting, placeholder handling, and a small set of duplicate-row reviews.

ACTION TYPE	STATUS	TARGET	REASON
cast_dtype	applied	aggregation_time → datetime64[ns]	Cast the column to inferred dtype datetime64[ns].
cast_dtype	applied	anno → Int64	Cast the column to inferred dtype Int64.
cast_dtype	applied	att_ivazioni → Int64	Cast the column to inferred dtype Int64.
cast_dtype	applied	attivazioni → Int64	Cast the column to inferred dtype Int64.
cast_dtype	applied	cessazioni → Int64	Cast the column to inferred dtype Int64.
rename_column	applied	aggregation-time → aggregation_time	Hyphenated name violated lowercase snake_case rules.
rename_column	applied	att ivazioni → att_ivazioni	Whitespace violated lowercase snake_case rules.
rename_column	applied	RATA → rata	Uppercase letters violated lowercase snake_case rules.
replace_placeholders_with_null	applied	descrizione_ente	Replace detected placeholder tokens with null values.
generate_cleaner	applied	RATA	Generate and apply a cleaner for pattern YYYY-MM.
drop_rows_candidate	proposed_not_applied	rows [255, 1709]	2 rows are exact duplicates after whitespace/case normalization.

Several applied actions aligned the dataset with consistent typing and naming conventions. For example, numeric measures such as **attivazioni** and **cessazioni** were cast to nullable integers, while text fields such as **descrizione_ente**, **fonte_dato**, and **_id** were standardized as strings. Placeholder replacement was also applied to fields including **codice_ente**, **provincia_sede**, and **regione_sede**, improving null handling without altering the underlying structure. Duplicate-row checks identified repeated records, but the listed candidates were only proposed and not applied, preserving the current row set. Overall, the remediation profile shows a strong emphasis on structural consistency and controlled normalization rather than aggressive row removal.

Verification Outcome

The verification outcome indicates that all identified column issues in **attivazioniCessazioni** were fully resolved. This means the dataset no longer contains the previously flagged column names, and the expected standardized names are now in place where applicable. In practical terms, the verification confirms that the dataset is aligned with the naming conventions required for downstream use.

COLUMN	RENAMED TO	STATUS	BEFORE	AFTER
mese	None	resolved	744	0
anno	None	resolved	585	0
attivazioni	None	resolved	414	0
cessazioni	None	resolved	413	0
RATA	rata	resolved	802	0
aggregation-time	aggregation_time	resolved	1607	0

All six issues were resolved, with the number of remaining occurrences reduced to zero in every case. Four columns were confirmed as resolved without renaming: **mese**, **anno**, **attivazioni**, and **cessazioni**. Two columns were renamed to standardized forms: **RATA** → **rata** and **aggregation-time** → **aggregation_time**. The before/after comparison shows a complete cleanup across the affected fields, with no unresolved instances left in the dataset.

Residual Risks & Manual Review Queue

- Exact duplicate pairs remain queued for confirmation before deletion.** The dataset **attivazioniCessazioni** contains multiple **drop_rows_candidate** items where two rows are exact duplicates after whitespace/case normalization. These include row pairs such as **[255, 1709]**, **[3109, 6669]**, **[3111, 13091]**, **[3799, 4442]**, **[3883, 7163]**, **[4039, 10953]**, **[4048, 8591]**, **[4134, 18098]**, **[4287, 10175]**, **[4549, 18731]**, **[4566, 6208]**, **[4663, 6851]**, **[4670, 8240]**, **[4676, 18634]**, **[4919, 18001]**, **[5070, 13158]**, **[5331, 19645]**, **[307, 11722]**, **[714, 12396]**, **[925, 17914]**, **[1105, 16882]**, **[1215, 12868]**, **[1568, 8435]**, and **[2263, 6276]**. These should be reviewed and removed only if the duplication is not intentional.
- Schema collisions across normalized column names require a manual merge decision.** Three duplicate-semantic conflicts are unresolved:
 - codice_ente** vs **CODICE ENTE** disagree on **57 rows** where both values are present.
 - provincia_sede** vs **Provincia Sede** disagree on **105 rows** where both values are present.
 - regione_sede** vs **regione%sede** disagree on **9708 rows** where both values are present. These conflicts should be resolved by selecting the authoritative source column, then backfilling or reconciling the alternate spelling consistently.
- Potential near-duplicate records need business-rule validation.** Two row pairs, **[160, 7699]** and **[841, 15282]**, share the inferred key columns **['_id', 'mese', 'anno', 'codice_ente']** but differ elsewhere. Because the key match is partial rather than exact, these records may represent legitimate updates or duplicated submissions; they should be checked against source logic before any consolidation.
- Recommended action order:**
 - Review and remove confirmed exact duplicates.
 - Decide the canonical form for the three conflicting column-name pairs.
 - Inspect the near-duplicate pairs to determine whether they are valid distinct records or accidental repeats.
 - Re-run validation after each change to confirm that no new conflicts are introduced.

Recommendations

- Prioritize review and resolution of the 56 deferred actions, especially those tied to schema, completeness, consistency, and cross-column findings.
- Investigate the 48 duplicate groups to determine whether they represent true duplicates, legitimate repeated records, or cases requiring deduplication rules.
- Address the 11 columns with missing values by confirming whether missingness is expected and, if not, defining targeted imputation or exclusion rules.

- 4. Review the 6 consistency findings and 8 cross-column findings to ensure business rules are fully encoded and to prevent recurrence in future loads.
- 5. Monitor the generated cleaners and verified fixes to confirm they remain stable in subsequent refreshes and do not introduce unintended transformations.

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